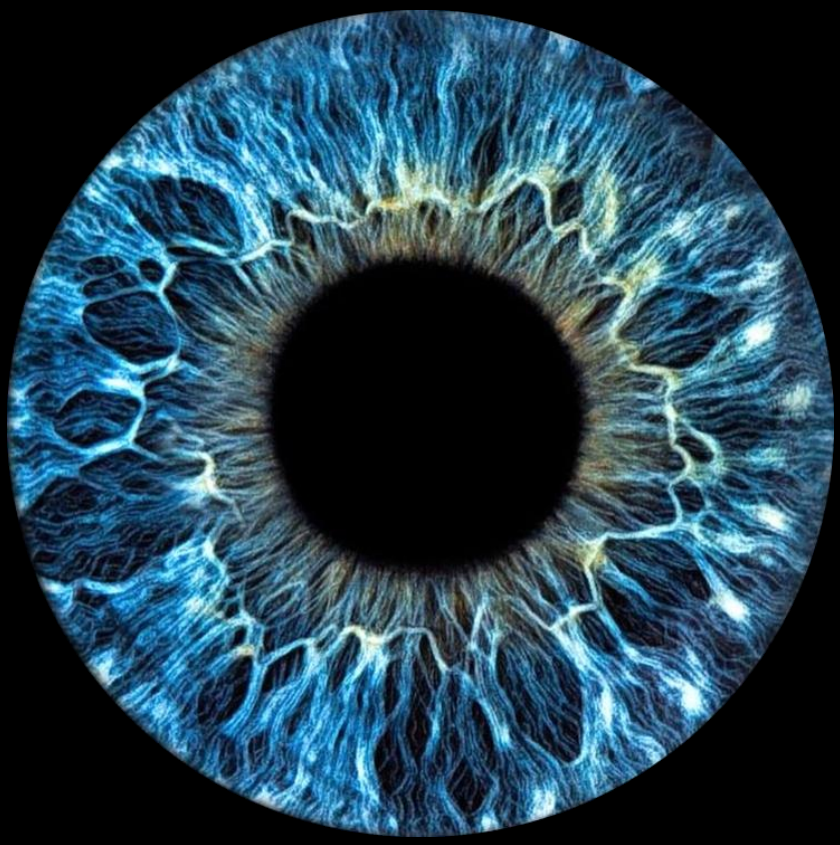
CORPORATE PROFILE

QblackAI

Simulate Your Vision

End-to-End Simulation Platform for Computer Vision



Simulate Your Vision

Product Video

CEO MESSAGE

Helping our customers make better decisions, earlier

Computer Vision redefined in photons

Cameras are no longer just recording devices. Robots, vehicles, and industrial systems rely on cameras to perceive the world and make human-like decisions. To build reliable vision systems, lenses, sensors, image processing, and AI algorithms must be designed together from the start.

Throughout my years in the industry, I saw firsthand how much potential is lost when each stage of a camera system is developed in isolation. Validating even a single configuration could require building a prototype, capturing images, and repeating the design process over and over.

At QblackAI, we aim to change how camera systems are designed and validated. We envision a process where every stage, from optics and sensors to image processing and AI algorithms, can be designed and evaluated together in simulation before the first prototype is built.

Our mission is to help teams building the next generation of vision systems move faster, make better-informed decisions, and achieve better results with fewer compromises.

QblackAI CEO

Derrick Kim

ABOUT US

From physical prototypes to Simulation

QblackAI is a software company that simulates the entire computer vision system based on physics. By reproducing the full imaging pipeline — from light source to AI algorithm — in a single environment, we help you find the optimal hardware·software combination before building a physical prototype.

MISSION

Reduce repeated prototyping in camera development and enable performance decisions at the design stage

VISION

In the era of Physical AI, become the standard platform for designing how machines see the world

COMPANY

Company	QblackAI Inc.
CEO	Derrick Kim
Founded	2026.02.11
Address	#539, 501 Teheran-ro, Gangnam-gu, Seoul, Korea
Business	Computer vision simulation software

THE CHALLENGE

Camera development relies heavily on physical prototypes

01 **Repeated prototyping**

Every lens·sensor·ISP combination must be validated by building samples and capturing images. Development time and costs increase with every additional combination.

02 **Late problem discovery**

The impact of optical aberrations and sensor noise on recognition only surfaces in field testing.

03 **Siloed development**

Optics, sensor, image processing, and AI teams each evaluate with their own metrics. It's hard to see how upstream choices affect downstream task performance.



What you need is not more prototypes,
but **evidence to decide before you build them**.

OUR SOLUTION

QBLACK

Physics-based camera simulation platform

From light source to AI algorithm, we integrate all six stages of a computer vision system into a single pipeline. Instantly see how changes at any stage affect overall system performance.

01	**Illuminant**	Standard illuminants & custom spectra
02	**Scene**	3D scene reconstruction from images
03	**Optics**	Lens aberrations & distortion
04	**Sensor**	Photoelectric conversion & RAW data
05	**ISP**	From RAW to processed final image
06	**AI Algorithm**	Detection·depth·segmentation metrics

— See sample outputs for each stage on the next page.

PIPELINE

From photon to perception

See how the same scene changes across six stages — the raw output of each simulation stage, exactly as it is.



01 Illuminant

Light Source



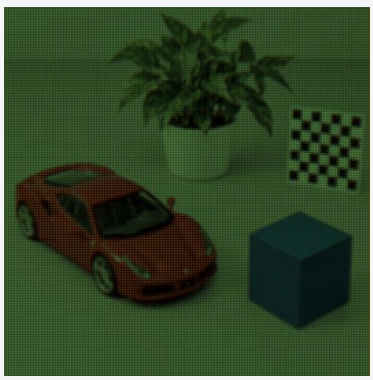
02 Scene

Scene Radiance



03 Optics

Lens



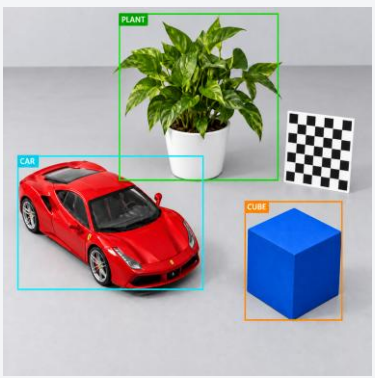
04 Sensor

Image Sensor



05 ISP

Image Signal Processing



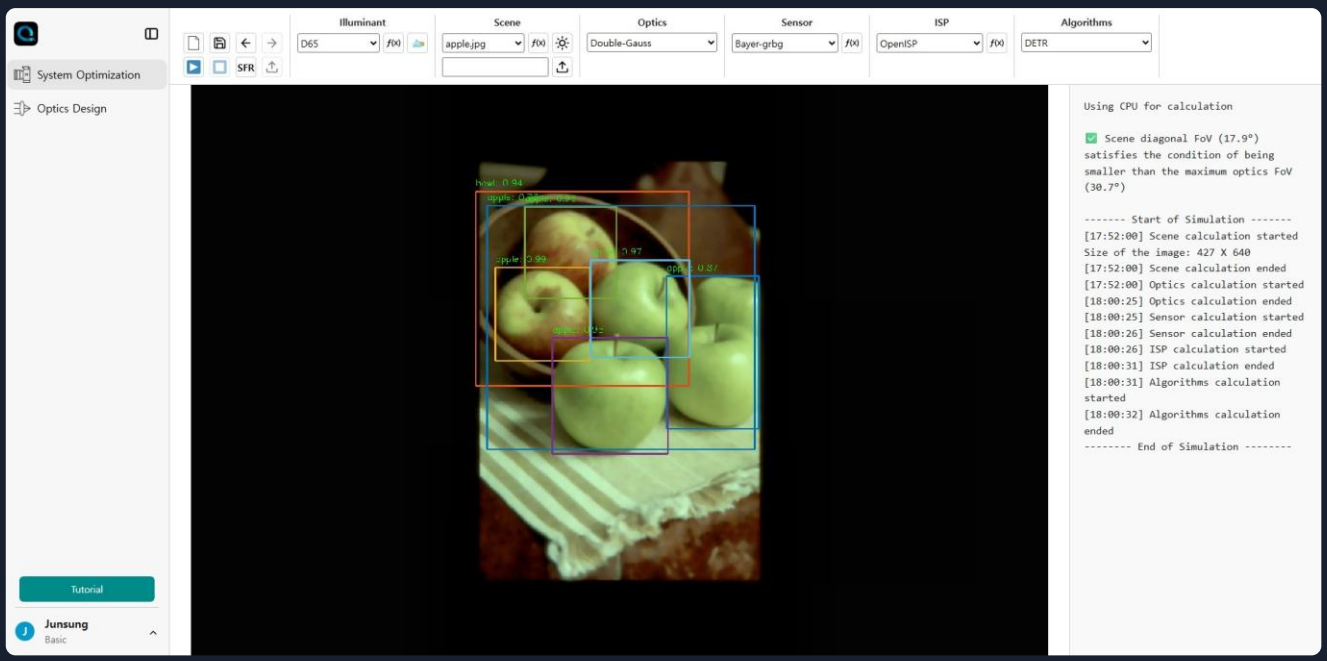
06 AI Algorithm

Perception

PRODUCT

Your spec sheet is all it takes
See everything on one screen

Enter lighting conditions, lens data and sensor specs to compare the final images and recognition metrics of candidate combinations on one screen.



Illuminant

Standard illuminants or custom spectra

Scene

Image-based scenes

Lens

Per-surface prescription — curvature, thickness, material, clear aperture

Sensor

Bayer pattern, quantum efficiency, pixel size, full-well capacity, PRNU/DSNU

Image Signal Processing

Demosaicing, white balance, color correction matrix, denoising, edge enhancement, gamma correction etc.

AI Algorithm

Object detection, depth estimation, segmentation algorithms

USE CASE 01

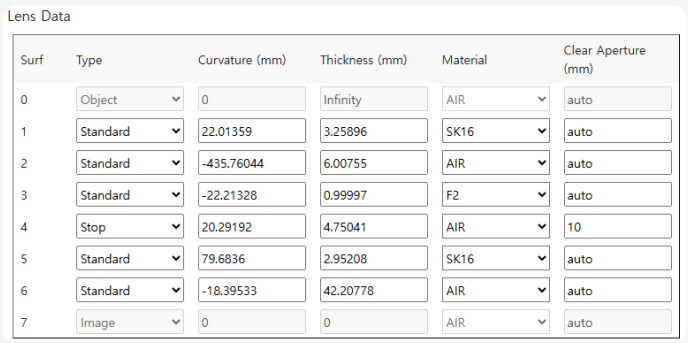
Comparing combinations before prototyping

"Compare combinations before you build them."

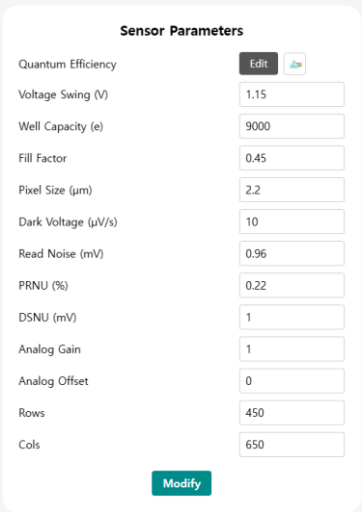
Your Pain Point

- Every lens, sensor and ISP combination requires physical prototyping and image capture
- Narrowing down candidates relies on individual engineers' experience
- The more combinations to evaluate, the greater the time and cost

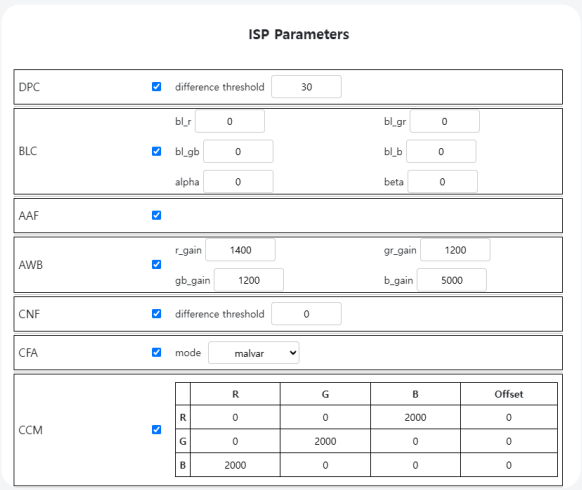
Enter your lens prescriptions, sensor and ISP specifications to compare final image quality and recognition performance across candidate combinations in a single environment. Each simulation takes 1–2 minutes on a GPU. Build physical prototypes only for the candidates that survive simulation.



Lens Input



Sensor Input



ISP Input

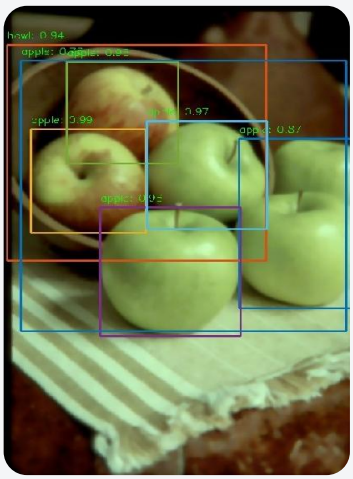
USE CASE 02

Pre-validating AI algorithms

"Know which camera your algorithm works best on, before you build it"

Your Pain Point

- The impact of camera HW on AI algorithm performance can only be validated in integration testing
- Algorithm and hardware teams evaluate with different metrics
- Low performance consistency, explainability, and reproducibility



See how changes in lens, sensor, and ISP specifications affect AI performance. Test how your AI will perform on real camera hardware before deployment, reducing the gap between training and real-world performance.

USE CASE 03

High-fidelity Training Data

"Synthetic data, as a real camera sees it."

Your Pain Point

- The gap between synthetic data and real camera footage makes trained models behave differently in the field
- Rendered images don't reflect the lens and sensor characteristics of the target camera
- The cost of capturing real data for every condition is expensive



Convert synthetic images into what the target camera's optics, sensor, and ISP would actually output, creating training data close to the real recognition environment. Narrow the gap between training and deployment where cameras drive decisions, such as robotics and autonomous driving.

INDUSTRIES

Every industry where cameras drive decisions

When cameras are designed for AI rather than human eyes, HW and SW must be optimized together.

INDUSTRY	YOUR PAIN POINT	WHAT WE DO
Mobile Imaging	Building and capturing images for every lens·sensor·ISP combination is repetitive and time-consuming	Find the best lens·sensor·ISP combination for image quality before prototyping
Autonomous Driving & ADAS	Changes in camera HW can significantly affect image quality and downstream AI performance.	Predict real-world performance before vehicle testing
Robotics & Humanoids	No clear criteria for choosing camera specs that fit vision algorithms	Select camera specs that fit robot vision algorithms
Industrial Machine Vision	Hard to predict how lighting, optics, or sensor changes affect inspection accuracy	Design lighting, optics, and sensors around inspection accuracy
AR·VR	Difficult to evaluate trade-offs between image quality and AI performance as cameras become smaller	Explore performance trade-offs under miniaturization constraints







HIGHLIGHTS

The numbers behind QBLACK

6 stages	One pipeline connecting everything from light source to AI algorithm.
1–2 min	Simulation time per combination on a GPU. (approx. 10 minutes on a CPU)
$0 fee	The software is free during joint validation PoC.

— Sign up at www.qblackai.com to try it for free.

HOW TO START

Built with our customers

QblackAI is an early-stage company, so we begin by validating our simulations against real hardware. Together, we use your specifications and equipment to verify accuracy and discuss potential adoption based on the results.

01

Free Trial

Sign up on our website and start right away. Explore the full pipeline using default light source, scene, optics, sensor, isp configurations.

02

PoC

We compare simulation against real measurements using your specs and equipment. Software is free. Test hardware is built together.

03

Subscription

Once validated, upgrade to subscription. PoC participants get 1 year free of charge.

PC PROCESS

01

Scoping

Define validation scope and evaluation criteria together.

02

Simulation

Enter HW specs and analyze candidate combinations.

03

Measurement

Build test hardware and measure with equipment.

04

Validation

Compare simulation and measurement results.

Thank You

QblackAI

Web	www.qblackai.com
Email	info@qblackai.com

Interested in free trial or joint validation PoC? Please email us.
We'll start by defining the validation scope and comparison criteria together.

Product Video

© 2026 QblackAI Inc. All Rights Reserved.